

TOPIC: Design and Development of a Remote-Controlled Drone for Safe and Efficient Betelnut Harvesting

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PROBLEM STATEMENT

Betelnut (areca nut) harvesting poses significant risks:

- Manual Climbing
- Skilled Labor Shortage
- Time and Cost
- Injury and Insurance

Current mechanical aids are inefficient, highly expensive and often damage trees or immature bunches.



IMPORTANCE OF THE PROJECT

This project addresses:

- Farmer Safety
- Precision Harvesting
- Better throughput
- Scalability
- Agricultural Modernization



PROPOSED SOLUTION

A remote-controlled drone(wired or wireless) equipped with:

- 1. Wireless Operation (Fully Autonomous Drone):** Current prototype is remote-controlled, but in the future the drone can be made wireless and autonomous with intelligent flight planning.
- 2. Automated Detection of Ripe Bunches:** With the integration of AI-based computer vision and machine learning, the drone can automatically detect ripe betelnut bunches based on color, size, and maturity indicators.
- 3. Remotely Controlled from Anywhere:** The drone can be connected via 4G/5G IoT modules to enable long-range remote control.
- 4. Enhanced Safety and Tree Protection:** Future versions can include advanced obstacle detection sensors and improved protective guards to ensure no damage to trees during harvesting.
- 5. Commercial Integration:** Once mature, the system can be offered as a service model (Drone-as-a-Service) where farmers lease drones during harvesting season.

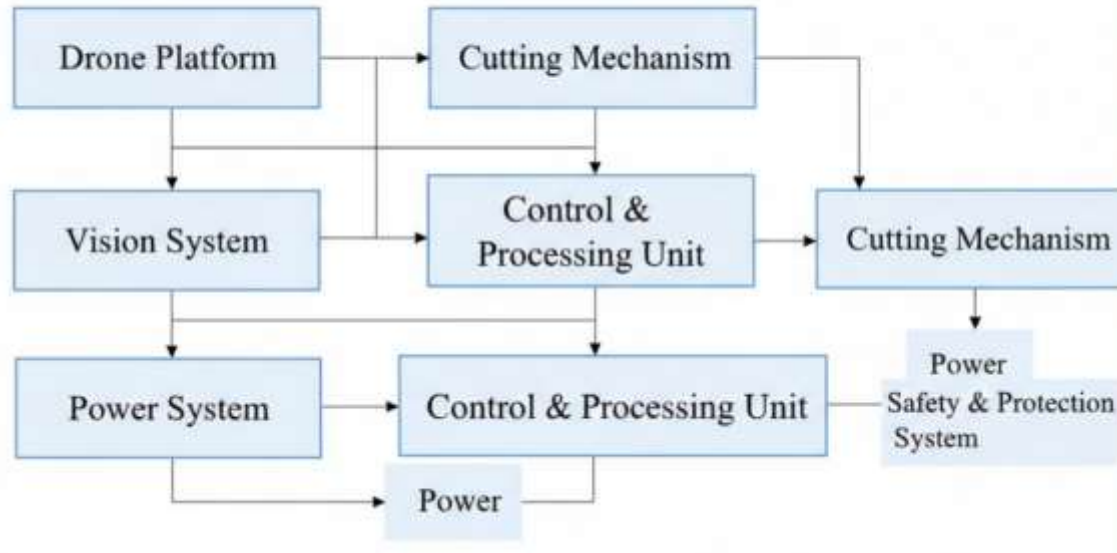
ADVANTAGES

- 5x faster harvesting
- Zero climbing risk
- On-demand harvesting service
- Slower long-term operating cost
- No climbing required
- Faster + safer + lower long-term cost
- Labour more easily available for drone control



SYSTEM ARCHITECTURE

Major Subsystems:



SYSTEM OPERATION

Camera Feed → Ground **Operator** / Controller



Flight Commands



Drone Flight Controller



Positioning Near Betelnut Crown



Cutter Arming & Actuation



Peduncle Shear Cut



Bunch Falls **to** Ground Net

CAMERA AND SENSOR SELECTION

Camera Type: Color, wide-angle (120°–150°)

Resolution: Minimum 1080p

Placement:

Bottom camera → navigation

Front/top camera → crown targeting

Additional Sensors(optional):

Time-of-Flight distance sensor

Limit switch inside cutter jaw

Current sensor for jam detection

EVALUATION & THROUGHPUT

Method	Trees/hour
Manual climbing	15–20
Pole cutter	25–30
Drone-based system	35–40

Evaluation Parameters:

- Safety improvement
- Reduction in labor dependency
- Time efficiency
- Tree health preservation
- Cost-effectiveness over seasons

Current Charges for Manual Labour:

Rs.13/ bunch and avg 2 bunches per tree
1 hr required for approx. 20 trees
1.45 kg/ tree (useful or processed)

CUTTER DESIGN

High-Force, Tree-Safe Cutting System:

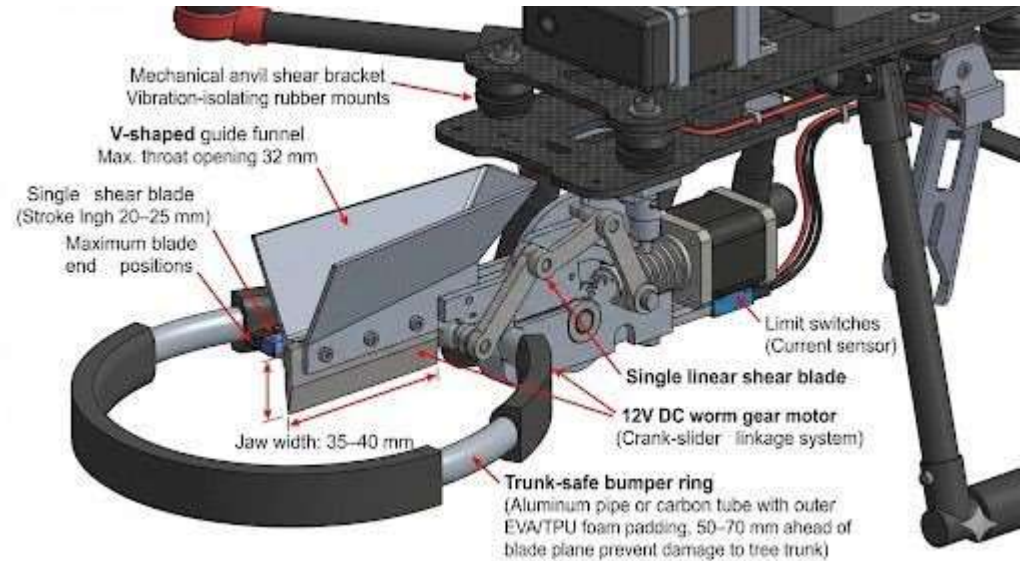
- Mechanical **anvil shear (non-rotary)**
- Designed for **2–3 cm peduncle**
- Single-stroke cut → low vibration
- V-shaped guide prevents slipping
- Physical diameter limiter
- Self-locking worm-gear actuation

Trunk-Safe Protection

- Foam-padded bumper ring
- Blade offset 50–70 mm from trunk
- Ring contacts trunk before blade
- No software dependency → purely mechanical safety
- Stable resting during cutting

Performance

- Peduncle diameter: 20–30 mm
- Design cutting force: ~2500 N
- Lever ratio: 5:1
- Motor torque required: ~20 N·m
- Achieved using low-cost DC worm gear motor



COMPONENTS REQUIRED

A. CUTTER MECHANISM (PRIMARY SYSTEM)

Component	Specification	Cost (₹)
Shear Blade	EN31 hardened steel (local machining)	2,000
Anvil Jaw	Mild steel, surface hardened	1,200
Pivot Shaft & Bush	Steel pin + brass bush	800
DC Worm Gear Motor	12V, 25 N-m torque	4,000
Crank & Linkage	MS fabricated (5:1 leverage)	1,800
Cutter Frame	Mild steel / aluminum	2,200

Subtotal (A): ₹12,000

B. SAFETY & STRUCTURE

Component	Purpose	Cost (₹)
Trunk-Safe Bumper Ring	Prevents trunk contact	1,500
EVA Foam Padding	Shock absorption	500
Propeller Guards	Rotor protection	1,500
Mounting Brackets	Drone integration	1,500

Subtotal (B): ₹5,000

C. CONTROL & ELECTRONICS

Component	Purpose	Cost (₹)
Motor Driver	30A H-bridge	1,200
Microcontroller	Arduino Nano clone	400
Limit Switches (2)	Blade position sensing	400
Current Sensor	Jam detection	600
Power Regulation	Buck converter	400
Wiring & Connectors	Harness & insulation	600

Subtotal (C): ₹3,600

D. CAMERA & VISUAL SYSTEM

Component	Specification	Cost (₹)
FPV Camera	HD wide-angle	2,500
Video Transmitter	5.8 GHz	1,500
FPV Receiver	Ground station	1,500
Display Monitor	5 inch LCD	2,000

Subtotal (D): ₹7,500

E. FABRICATION & MISCELLANEOUS

Item	Purpose	Cost (₹)
Machining & Welding	Blade, frame, mounts	4,000
Fasteners & Hardware	Nuts, bolts, spacers	1,200
Paint & Protection	Corrosion resistance	400
Subtotal (E): ₹5,600		
GRAND TOTAL		₹33,700

PROJECT TIMELINE

Activity	Duration
Design & calculations	Ready
Cutter fabrication	2 weeks
Drone integration	1 week
Software setup	1 week
Local testing	2 weeks
Field trial prep	1 week

Growth Strategy

- Partner with agri-cooperatives and FPOs
- Service model for rapid rural penetration
- Demonstration-based sales events
- Government agri-mechanization subsidies
- Build drone fleet leasing network via franchise operators

BUSINESS MODEL

Total Revenue Potential

India has:

- 7-8 Lakhs areca-nut farmers
- 650,000+ hectares of plantations
- Annual market value: \$1.4B+
- Converting just 1% of farms → \$50M+ revenue opportunity. Scaling across SE Asia significantly expands TAM.

Business Model

- Direct Drone Sales to farmers
- Pay-per-Tree Harvesting Service model
- Annual Maintenance & Battery Subscription
- Fleet Leasing for Agri-contractors
- Customers save 40–60% in labor cost, increase harvesting speed, and avoid crop loss from delayed picking.

3-5 Year Outlook:

- We will be initially offering a DaaS model: Drone as a Service
- We will charge 25 per tree so an average plantation of say 500 trees will make 12500 per day
- Extrapolating that to a month will be 12500 X 20 That is 2.5L per drone
- Assuming 20 plantations per town and one drone for that, yearly revenue will be 30L per town
- Estimated revenue for first year for one state, assuming 100 drones in service: 30Cr

SUMMARY

- Develop a **drone-based betelnut harvesting system** to overcome challenges of manual harvesting such as **labor risk, time consumption, and low efficiency**.
- Integrate a **quadcopter platform with GPS stabilization** for precise and stable flight.
- Use **rotary/scissor cutting mechanisms** along with **net protectors** to ensure **safe, accurate, and eco-friendly harvesting** without damaging the tree.
- **Eliminate the need for manual climbing**, significantly improving **farmer safety**.
- **Increase harvesting speed and efficiency** while maintaining crop quality.
- Designed with a **clear phased development plan** and **feasible budget** for practical implementation.
- Scalable for **future upgrades** such as **wireless operation, AI-based ripe bunch detection, and 4G/5G remote control**.
- Combines **modern drone technology with agricultural needs** to offer a **sustainable, farmer-friendly, and scalable solution** for plantation harvesting.

REFERENCES

ICAR–CPCRI (Arecanut cultivation & harvesting)

Govt. of India – Agri Drone Mission

DGCA India – Drone operation guidelines

Shigley's Mechanical Engineering Design

DJI Enterprise Payload Integration Docs

Indian electronics & fabrication vendors